Introduction to Software Engineering

Project Idea & Team Contract



Software Engineering Department
Faculty of Information and Technology
University of Science

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Project Idea & Team Contract

Objectives

This document focus on the following topics:

- ✔ Describing the project idea at a high level so the teaching staff can approve it
- ✔ Agreeing on how the team will work together (team contract)
- ✔ PA0 is not graded, but it is required before the team starts PA1

1 Team Information

Item	Value
Team ID	10
Team name	DiDung
Team lead	Truong Le Trung Hieu
Communication channel	Discord

Student ID	Full name	Email	Main role	Backup role
24127169	Truong Le Trung Hieu	tlthieu2428@clc.fitus.edu.vn	Team lead / PM	
24127130	Tran Minh Tien	tmtien2418@clc.fitus.edu.vn	Business analyst	Developer
24127169	Truong Le Trung Hieu	tlthieu2428@clc.fitus.edu.vn	Developer	Tester
24127035	Vo Dinh Cao Minh Hao	vdcmhao2420@clc.fitus.edu.vn	Tester	Developer
24127020	Nguyen Thanh Dat	ntdat2428@clc.fitus.edu.vn	Designer	Developer

2 Project Idea Overview

2.1 Project Name and One-Sentence Pitch

DiDung – a web app that helps people in Ho Chi Minh City decide when and where to go out, by monitoring fine dust (PM2.5/AQI), rain and high-tide flooding, forecasting the next few hours, and warning you when conditions cross your own thresholds.

2.2 Problem and Motivation

Residents of Ho Chi Minh City face three everyday risks that often happen together:

- **Fine dust (PM2.5):** Air pollution regularly reaches levels that are unhealthy, especially for children, the elderly and people with respiratory conditions.
- **Heavy rain:** Sudden downpours disrupt commuting and outdoor work.
- **High-tide flooding:** Many streets in the city are flooded when the tide peaks, and rain at the same time makes it worse. People often find out only when they reach a flooded road.

Today, people check separate apps, news pages or word of mouth for each risk, usually only at city level. Most sources show current readings rather than short-term forecasts, and none lets users set personal alert thresholds.

DiDung brings AQI, rain and tide/flood risk into one place and forecasts them a few hours ahead. Users choose their district and their own thresholds, and the system notifies them when a forecast crosses those thresholds, so they can decide when to go out and which route to take.

2.3 Target Users

User group	What they need from the system
Residents and commuters	A quick view of current and upcoming AQI, rain and flood risk in their area, plus simple advice on whether and when to go out
Residents in flood-prone areas	Early warning before high tide combined with rain, so they can move vehicles or avoid affected streets

Outdoor workers and athletes (delivery drivers, runners, etc.)	Short-term forecasts of rain, air quality and flood risk to plan trips, routes and training
System administrator	Visibility into data collection, errors and forecast accuracy, to keep the service reliable
Sensitive groups (children, elderly, people with respiratory conditions) and their families	Alerts at a lower, personally chosen AQI threshold, so they can avoid exposure to fine dust

2.4 Main Features

#	Feature	Short description	Must-have?
1	Data collection and storage	Scheduled hourly jobs fetch PM2.5/AQI from OpenAQ (fallback: Open-Meteo Air Quality), rain/weather from Open-Meteo, and tide levels from published tide tables or monitoring stations. A separate periodic job collects flood reports from online news (RSS feeds of Vietnamese news sites) to build a list of known flood hotspots. Data is checked for gaps or anomalies, then stored in a database.	Yes
2	Short-term forecast	Provides AQI and rainfall forecasts for the next few hours using available API forecasts or a simple baseline.	Yes
3	Basic ML model	Trains a simple model (e.g., XGBoost/LightGBM), compares it with the baseline using time-based evaluation, and uses it if it performs better.	No
4	Tide and flood-hotspot warning	Forecasts the tide peak for the next hours and combines it with the rain forecast to give a flood-risk level (low / medium / high) per district. When the risk is high, the known flood-prone streets in that district, collected from news reports and city announcements, are highlighted on the map as “at risk”. The system warns about areas and known hotspots, and does not predict flooding on individual streets.	Yes
5	User accounts and regional alerts	Users register and log in, choose their districts and set personal thresholds (AQI, rain, tide level).	Yes
6	Alert banner and notification center	When a forecast exceeds a user’s threshold, a banner is shown and the alert is saved in a notification center.	Yes

7	Dashboard	Station-based AQI map, flood-risk areas, and history and forecast charts for the selected area.	Yes
8	Advices	A short recommendation on whether it is a good day to go out, based on the AQI, rain and flood-risk forecasts.	No
9	Admin monitoring page	Logs each data fetch, reports errors when data stops arriving, and charts forecast errors over time.	No
10	Mobile push notifications	Push alerts on mobile, if time allows.	No
11	User flood reports	Logged-in users can mark a street as "flooded now". Reports are shown on the map and used to update the hotspot list and check how accurate the warnings are.	No

2.5 Similar Products

- **Weather apps (Google Weather, AccuWeather, Windy, and the national hydro-meteorological forecasting website nchmf.gov.vn):** They provide rain and weather forecasts, and the national site also publishes tide and flood information. However, they are general weather tools. They do not show air quality, flood-prone streets, or per-user thresholds. Windy's API is also paid, so we do not use it as a data source.
- **Waze and Google Maps (traffic and route apps):** They are used for navigation, and Waze lets users report road hazards such as flooding. These reports depend on drivers happening to submit them, there is no forecast, and there is no air quality information. DiDung warns in advance using tide and rain forecasts, and gives advice on whether and when to go out.
- **Ho Chi Minh City flood-control information channels (city websites, news sites, social media groups):** They publish flood locations and warnings, but the information is scattered and mostly reactive. DiDung collects known flood hotspots from news into one map and links them to a forecast risk level.
- **Windy:** It has excellent weather visualization, but it is a general-purpose tool, and its API is paid, so we don't use it. DiDung is simpler and focuses on the three local problems of one city.

2.6 Platform and Technology Ideas

Platform: Web application (responsive, so it is usable on a phone browser). A native mobile app is out of scope.

Technology ideas (not final):

- **Backend:** Python (FastAPI or Django)
- **Frontend:** React (or plain HTML/JS if the team prefers), Leaflet for the map, Chart.js for charts
- **Database:** PostgreSQL
- **Scheduled jobs:** APScheduler or cron: hourly data collection, hourly forecast, periodic news collection
- **Forecasting:** scikit-learn / XGBoost / LightGBM
- **Datasources:** OpenAQ v3 (free API key) and Open-Meteo (no key) for air quality and weather; published tide tables or monitoring-station data for tides; RSS feeds of Vietnamese news sites for flood hotspots
- **Hosting:** Render, Railway

2.7 Feasibility, Scope and Open Questions

Feasibility. We believe a team of 4 students can build the Must-have features in about three weeks of coding, but the schedule is tight and depends on keeping the scope small. A possible split: one person on data collection and storage, one on forecasting, one on accounts and alerts, one or two on the map and dashboard. We will use simple models and build the data pipeline first, because the forecast, the alerts and the dashboard all depend on it.

Out of scope:

- Native mobile app and mobile push notifications
- Predicting flooding on individual streets (we only warn by district and known hotspots)

- User flood reports and the admin monitoring page (nice-to-have, only if time remains)
- Medical advice (daily advice is only general guidance based on AQI, rain and flood risk)
- Coverage outside Ho Chi Minh City

Main risks:

1. **Data availability:** OpenAQ may have few stations in HCMC, and some stations may have gaps. Mitigation: fall back to Open-Meteo Air Quality and use interpolation for districts without a station.
2. **Tide data:** We have not confirmed a free, reliable API for tide levels on the Saigon River. Mitigation: use published tide tables loaded into the database.
3. **Flood hotspots from news:** Extracting street names automatically is error-prone. Mitigation: semi-automatic collection, with a manually reviewed starting list of about 20–30 known hotspots.
4. **Forecast quality:** We may have little collected history for training. Mitigation: backfill with historical data from the APIs, and keep the baseline as a fallback if the ML model does not beat it.
5. **Schedule:** Six Must-have features in three weeks is ambitious. Mitigation: build in order of dependency (data, forecast, dashboard, accounts and alerts, tide/flood) and cut the tide/flood feature to a simple version if we fall behind.

Questions for the TA:

1. Is it acceptable to collect flood hotspots semi-automatically (script gathers news links, team reviews them) instead of fully automatically?
2. For "fully implemented" forecasting, is a simple baseline plus one ML model enough, and how should we evaluate it (e.g., MAE against the baseline)?
3. Is in-app notification (banner and notification center) enough, or is email also expected?
4. Can we preload a reviewed list of flood hotspots and historical data for the demo?
5. Is the Must-have list too large for three weeks? If so, which feature should we reduce first?

3 Team Contract

3.1 Goals and Expectations

Our team aims to achieve a grade of 9–10 out of 10. Each member commits to the weekly working hours listed below:

Roles	Before week 8	Weeks 8–10	Main tasks
Team lead / PM	5–6 hours/week	8–10 hours/week	Manage scope, dependencies and progress; communicate with the TA; coordinate integration; prepare weekly reports and demos; prioritize essential features when the schedule is tight.
Business analyst	6–8 hours/week	8–10 hours/week	Finalize requirements and acceptance criteria; assess AQI, weather and tide data availability; define district coverage, personal thresholds and flood-risk rules; manually review flood hotspots collected from news.
Designer	6–8 hours/week	8–10 hours/week	Design the responsive dashboard, map layers, forecast charts and notification center; clearly distinguish measured data, forecasts and flood-risk estimates; support frontend development and review interface usability.
Developer	6–8 hours/week	10–12 hours/week	Build data collectors, database and APIs; import tide tables and reviewed flood hotspots; implement baseline and ML forecasting, district flood-risk rules, accounts

			and alerts; integrate the dashboard and deploy the application.
Tester	5–6 hours/week	8–10 hours/week	Prepare test cases and sample data; verify data freshness, district mapping and hotspot accuracy; compare forecasts with baselines; test alert thresholds, missing data, API failures and responsive layouts; support acceptance testing.

3.2 Communication

✔ Main channel for discussion: ***Discord (Join link)***

✔ Expected response time to messages: *within 24 hours on weekdays, within 12 hours on weekend*

✔ Where decisions and files are kept:

- **Task board:** Link Jira
- **Link repository:** Link Github
- **Link shared drive**: *Nhập môn công nghệ phần mềm*

3.3 Meetings

✔ Weekly status meeting with the TA: To be confirmed with TA

✔ Internal team meetings:

- **Thursday:** 20:00 every week (Online - Google Meet)
- **Sunday:** 20:00 every week (Online - Google Meet)

✔ Attendance rule: inform the team in advance if you cannot attend; missed meetings are recorded in the weekly status report

✔ Weekly status report deadline chosen by the team: **Sunday** - 23:59 every week.

3.4 Roles and Responsibilities

Responsibility	Main	Backup
Project planning, scheduling & TA communication	PM	BA
Requirements analysis & documentation	BA	PM
UI/UX design & frontend interface	Designer	Developer
Backend, database & data pipeline development	Developer	Tester
QA: test planning & test case design	Tester	BA
QC: test execution, bug reporting & regression testing	Tester	BA
Bug fixing & code review	Developer	Tester
CI/CD & deployment	Developer	PM
Weekly status reports	BA	PM

AI Usage Log	BA	PM
Presentation coordination	PM	Designer

3.5 Work Standards

✔ **Internal deadline:** Must have done the work before the deadline 24 hours for team review.

✔ **Quality:**

- Every document section must be reviewed by at least one other team member before final submission.
- Every code change must be submitted through a pull request and reviewed by at least one other member before merging.
- Direct pushes to the protected main branch are not permitted.
- Code should follow consistent naming conventions, formatting, and project structure.
- Relevant tests should be completed successfully before a task is considered finished.

✔ **Tools:**

- Every task must have a clear description, assigned owner, due date, and progress status.
- A task is considered completed when its deliverables are finished, reviewed where applicable, and updated on the task board.
- Members who anticipate missing a deadline must inform the team as early as possible and propose a recovery plan.

3.6 Use of AI Tools

- ✔ **Tools:** ChatGPT/Codex, Claude
- ✔ **Purpose:**
  - Brainstorming project ideas and possible solutions.
  - Assisting with technical research and problem analysis.
  - Supporting code development, debugging, and test case generation.
  - Reviewing documentation for clarity, consistency, and language quality.
- ✔ **Verification and Review:**
  - All AI-generated outputs must be reviewed and verified by the responsible team member before use.
  - AI-generated code must be understood, tested, and modified when necessary.
  - AI-generated documentation must be checked for accuracy and consistency with project requirements.
  - Every member must be able to explain the work they submit.
  - Personal or confidential information must not be provided to AI tools.
- ✔ **AI Usage Documentation:**
  - Each member is responsible for reporting significant AI usage, including the tool used, task or purpose, and how the results were verified or modified.
  - The Team Lead will consolidate the AI Usage Log for each PA submission, with the Business Analyst serving as backup.
  - All AI usage must comply with the course's Generative AI policy.

3.7 Decision Making

The team will prioritize discussion and consensus when making project-related decisions.

If consensus cannot be reached, the team will use a majority vote, with at least three of four members participating.

In the event of a tie, the Team Lead will make the final decision after considering the opinions of the involved members.

Major decisions involving project scope, technical architecture, deadlines, or changes to agreed deliverables must be discussed with the team and documented.

Significant changes affecting approved project requirements will be communicated to the TA when necessary.

3.8 Handling Conflicts and Missed Work

✔ **Step 1 — Direct Communication**

- When a member misses a deadline or fails to fulfill an assigned responsibility, the Team Lead or relevant member will discuss the issue directly with the person involved.
- The team will identify the cause, agree on corrective actions, and establish a revised completion date.

✔ **Step 2 — Team Discussion**

- If the issue remains unresolved or occurs repeatedly, it will be discussed during a team meeting.
- The discussion, agreed actions, and any task reassignment will be recorded in the weekly status report.

✔ **Step 3 — TA Involvement**

- If the issue cannot be resolved internally or significantly affects project progress, the Team Lead will inform the TA and provide relevant evidence.
- Such issues may be considered by the teaching staff when evaluating individual contributions.
- All members agree to handle conflicts respectfully, communicate transparently, and prioritize project completion.

3.9 Signatures

Student ID	Full name	Signature	Date
24127169	Truong Le Trung Hieu		12/10/2026
24127020	Nguyen Thanh Dat		12/10/2026
24127130	Tran Minh Tien		12/10/2026
24127035	Vo Dinh Cao Minh Hao		12/10/2026

4 AI Usage Log

#	Member	Tool	Task / purpose	How the output was verified and changed
1	Nguyen Thanh Dat	*ChatGPT*	*Complete Team Contract*	Reviewed AI-generated suggestions against the PA0 template and course guidelines; adapted team roles, communication rules and deadlines to the team's actual agreements.